

Democratizing Data + AI



Agenda

- Introductions and Lecture Content – 45 minutes
- Hands on Ingestion and Dashboarding – 45 minutes
- Q&A – 30 minutes



Introductions

- Leigh Robertson
 - UVA 2016 Systems and Information Engineering
 - Career
 - CapTech Consulting 2016 – 2021
 - Lead Data Engineer – Innocean 2021 – 2022
 - Senior Data Engineer – Vizio 2022–2024
 - Solutions Architect – Databricks 2024–Present
 - Emerging Enterprise 2024– Feb 2026
 - DNB Strategic Feb 2026 – Present
 - Current Location: Denver, CO
 - Interests:
 - Skiing/ snowboarding
 - Hiking
 - UVA basketball



Hello & Welcome!

About Your Instructors



```
{  
  name: "amanda laforest",  
  location: "denver, co",  
  role: "sr. delivery solutions engineer",  
  background: ["data_science", "ai_ml"],  
  schools: ["umich", "uva"],    # go hoos!  
  social: "linkedin.com/in/amalaforest/",  
  hobbies: ["running", "audiobooks"],  
  funfact: tellUsAFact()  
}
```

Introductions

- Elaine Adler
 - SUNY Buffalo / TCU
 - Career
 - Guidehouse
 - Tableau / Salesforce
 - Databricks
 - Current Location: Charlottesville, VA
 - Interests:
 - Hiking, Yoga, Traveling, Baking...
although time is extra limited these days!



Introductions

- Tom Seufert
 - GMU
 - Career
 - Pivotal/VMware
 - ASAPP
 - Databricks
 - Current Location:
Charlottesville, VA



10,000+
global customers

Inventor of the **lakehouse**
&
Pioneer of **generative AI**

\$5.4B+
ARR



databricks

The data and AI
company

Gartner-recognized Leader

Database Management Systems
Data Science and Machine Learning Platforms

Forrester Wave Leader

Data Lakehouses
AI Foundational Models for Language

\$14B
in investment

Creator of



Mosaic MPT

DBRX



The Architects of the Modern Data Stack

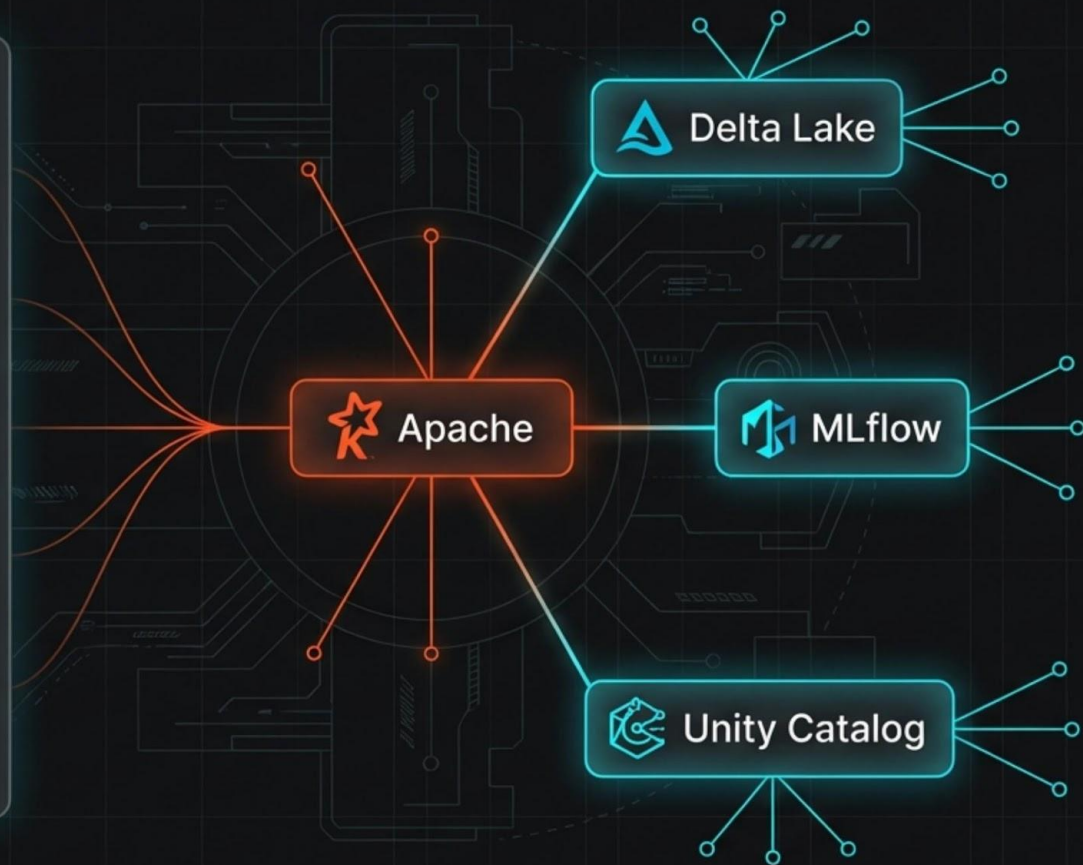
Founded by 7 PhDs from
U.C. Berkeley (AMPLab)

Creators of Apache Spark

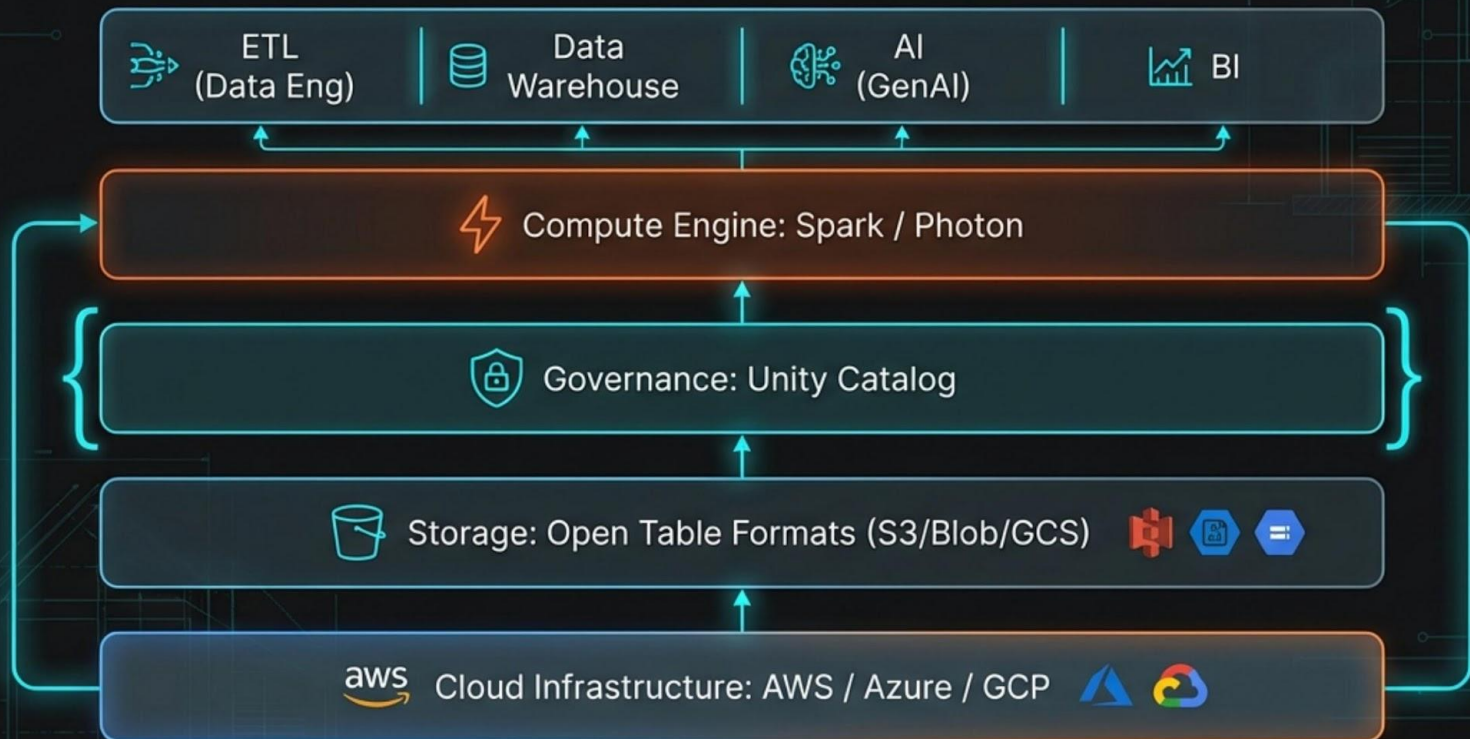
\$14B Investment

\$5.4 B ARR

10,000+ Global Customers



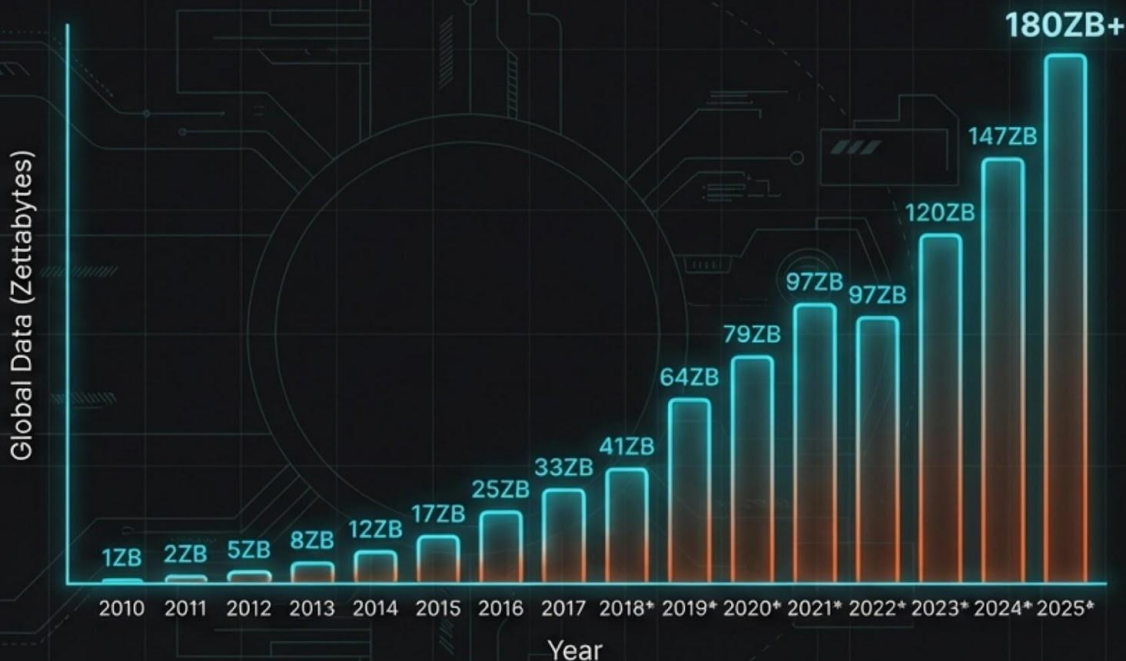
The Platform Architecture



The Data Explosion Broke the Old Architecture

The Goal: Access, Quality, Freshness.

- **The Reality:** Data Silos, Security Gaps, Complexity.



Bad Data = No AI.

Era 1: The Data Warehouse (1980s - 2000s)

Structure: Schema-on-write (Star Schema)



ETL

JetBrains Mono



DATA WAREHOUSE

Inter Regular



BI



Reports

Use Case: BI & SQL Reporting

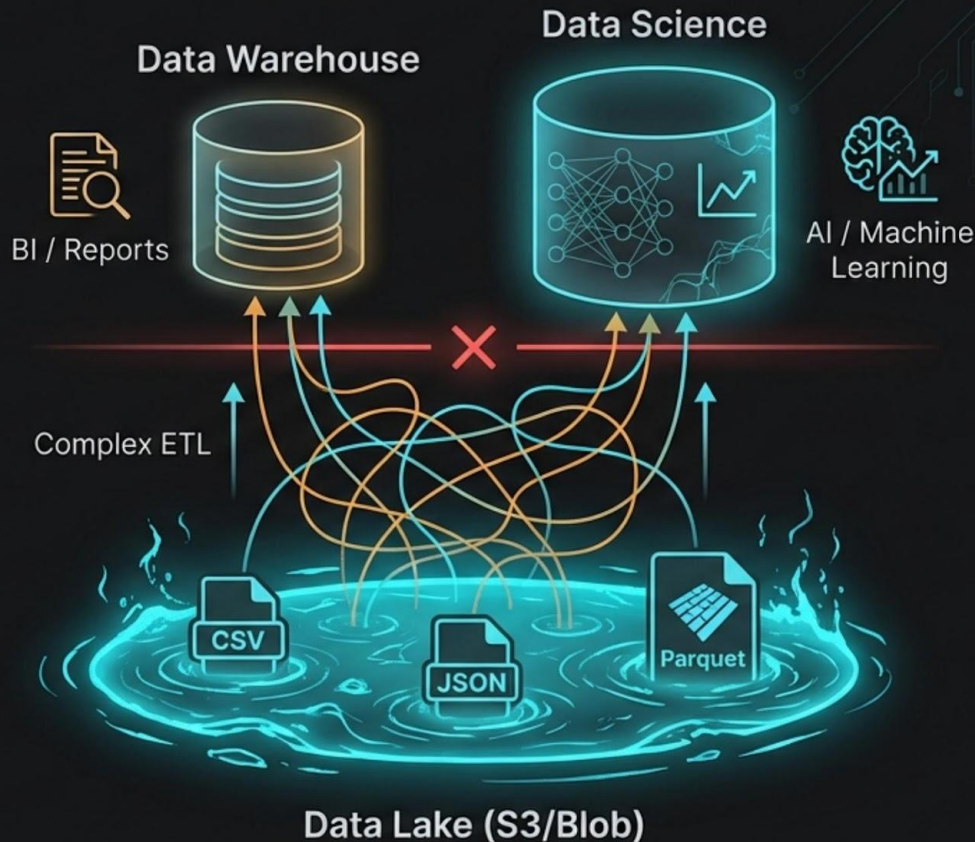
Limitation: Expensive (\$\$),
No Unstructured Data Support



AI / Machine Learning?

Era 2: The Data Lake (2010s)

- ✓ Cheap Storage (Object Store)
- ✓ Open File Formats
- ✓ The Failure: Data Silos & Governance Chaos
- ✓ Result: Poor Reliability



The Convergence: The Data Lakehouse



Eliminates the two-tier architecture.
All data (structured & unstructured) in one place.

Three Tenets of the Lakehouse

Open Formats & API



Open Formats & API

No vendor lock-in. Data stored in open formats.

JetBrains Mono

Delta Lake / Iceberg

Unified Governance



Unified Governance

Security wraps the whole platform, not just the warehouse.

JetBrains Mono

Unity Catalog

Performance



Performance

Warehouse speeds on Data Lake storage.

JetBrains Mono

Photon Engine

Anatomy of Storage: Delta Lake

From File Abstraction to Table Abstraction



The **Delta Log** tracks state: **ACID Transactions**, **Schema Enforcement**, **Time Travel**.

Anatomy of Governance: Unity Catalog

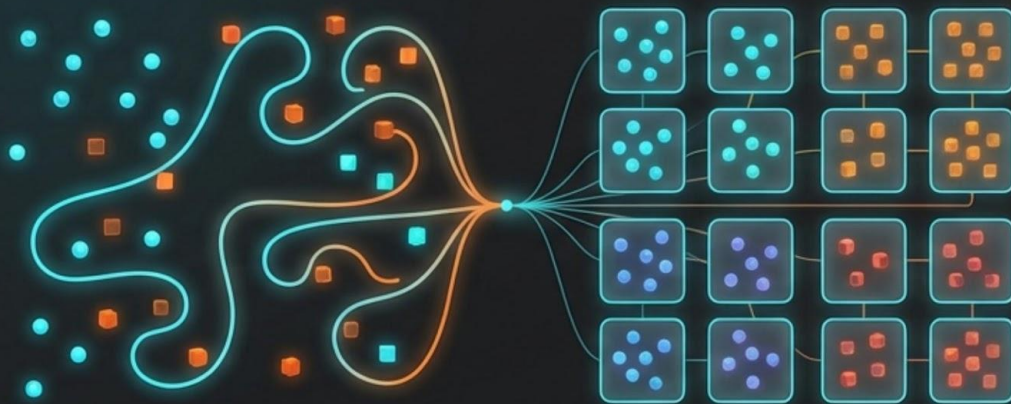


Intelligent Optimization: 'No Knobs'

The Problem: Manual Tuning



Liquid Clustering



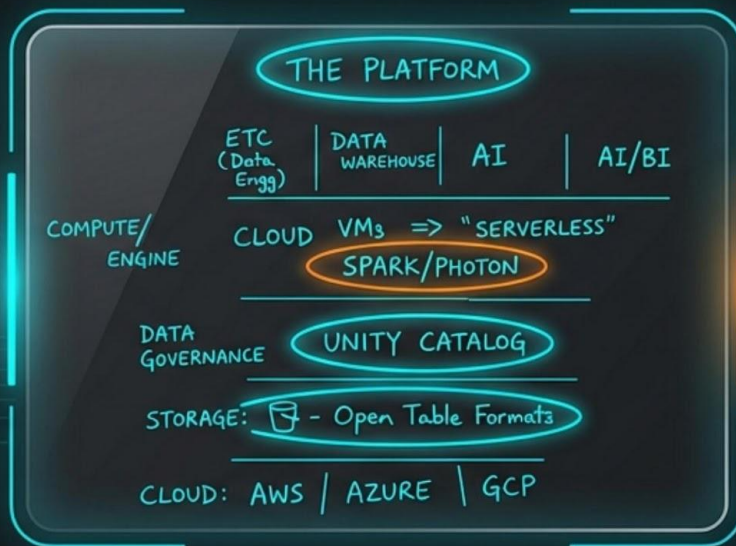
- **Automated Features**

- Auto-Optimized Infrastructure (**Serverless**)
- Predictive I/O (**Intelligent Data Skipping**)
- Liquid Clustering (**No Manual Partitioning**)

A Unified Future

SIMPLE

(One platform for
Warehouse,
Engineering, AI)



OPEN

(No proprietary lock-in,
Open Formats)

MULTI-CLOUD

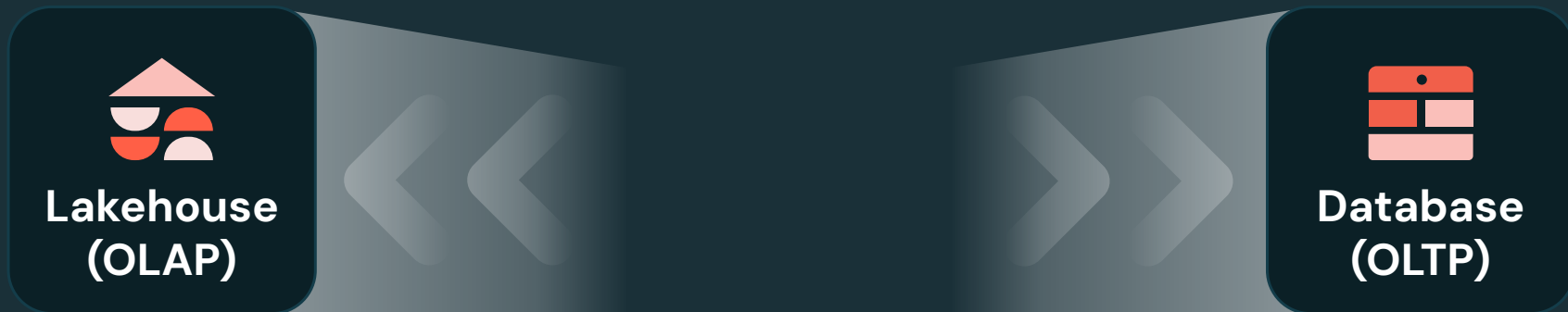
(AWS / Azure / GCP)

The architecture that turns raw data into Data Intelligence.

Analytics and
operational use
cases are **starting**
to **converge**

AI-assisted
applications are
accelerating
the trend

Operational and analytical systems must work together



But right now they're in separate worlds



The architectural tax



You need a modern way to manage all of your data



**Lakehouse
(OLAP)**



**Database
(OLTP)**

You need a Lakebase



Lakebase: Your database for the AI era



Apps

Agents



Lakebas
e



Unity Catalog



DELTA LAKE

ICEBERG



Why OLTP Databases at Databricks?

New class of applications require both transactional and analytical data

- Faster decision making to improve products and/or manage costs
- Example use cases:
 - E-commerce: pre-calculated customer segmentation and analytics are used to drive workflow e.g. preferential delivery, offer targeting, product recommendations
 - Healthcare: Clinical workflow managing recommendations and workflow management of clinical trial data
 - FinServ: automatic market trading
 - Retail: chatbot that uses recent chat history to pass into the context window and works on real-time application data e.g. shopping cart
- Analytics and operational systems are starting to converge
- AI assisted applications are accelerating the trend

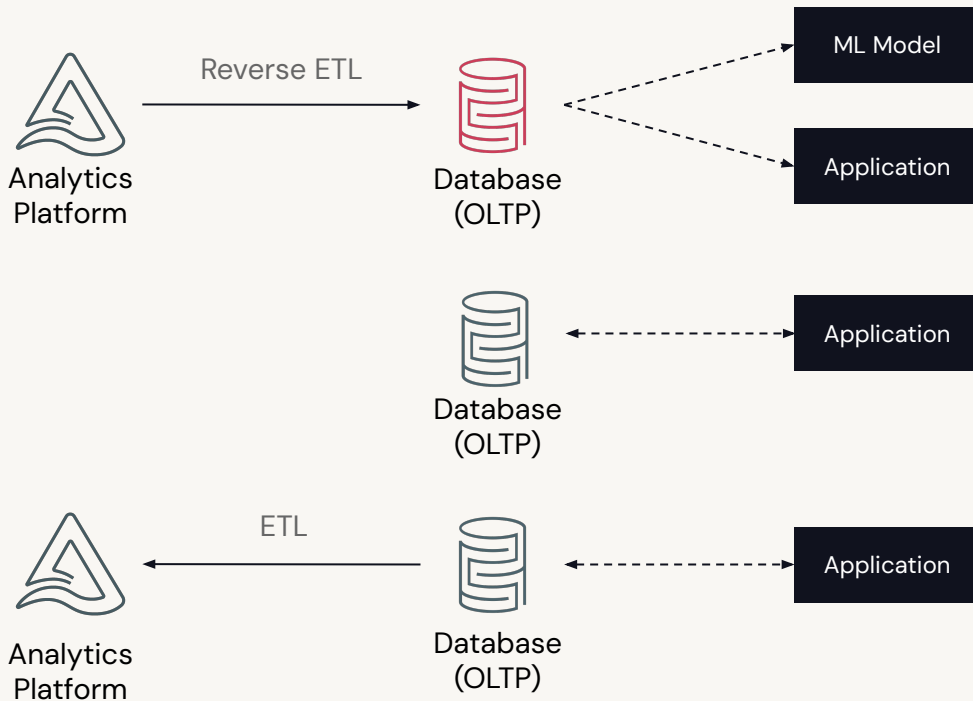
Workload Types

1. Online Feature Store

2. Data Serving to applications

3. Storing application state

4. Analytics over transactional data



Lakebase is...

A Postgres Transactional Database

that is

- ↗ Fully-managed
- ↗ Integrated with the Lakehouse
- ↗ Developer friendly

Terminology

- Classic compute
 - Use customer instances from their cloud environment
- Serverless compute
 - Databricks provided compute
- Medallion Architecture
 - ETL framework for ingesting data
- Pricing
 - Everything is tied to DBUs



Databricks Components

Older

ETL

- Spark declarative pipelines
- Procedural ETL

Orchestration

- Workflows

Warehousing

- SQL warehouses

Exploration

- Notebooks

Dashboarding

- AI/BI

Newer

ETL

- Lakeflow connect

Gen AI

- Model serving

OLTP

- Lakebase

Message Bus

- Zerobus

